High-Precision Computational Tests on ζ(3) and π

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Abstract

Methodological contribution. We identify and formalize a general reproducibility pitfall in PSLQ-based integer-relation searches: raising the target bound (`maxcoeff`) without also raising the iteration budget (`maxsteps` , which defaults to 100 in mpmath) leaves the algorithm returning `None` whether it has genuinely certified non-existence or has merely run out of iterations - the two are indistinguishable without inspecting the algorithm's internal termination state. We quantify how sharply this degrades with basis size (a rate of ~0.19 certified decimal digits per iteration on a 3-element basis collapses to ~0.0004 on a 26-element basis and to an outright 0 on a 28-element basis, all under identical settings), show that an earlier draft of this manuscript fell into exactly this trap on three separate tests, and demonstrate the fix: independent cross-validation via LLL lattice reduction (fpylll), which produced certificates in seconds where the trapped PSLQ runs needed 8.7-12.3 hours to produce a near-worthless bound. We believe this diagnostic is of more general use than any specific bound in this paper.

Numerical results, using two independent integer-relation engines - mpmath's PSLQ and LLL via fpylll: (1) No relation $a \cdot \zeta(3) + b \cdot \pi^2 + c = 0$ exists with coefficient norm $\leq 10^{19999}$ (LLL, 60000-digit scaling), independently confirmed by PSLQ to 10^{18695} (40000 digits, 100000 iterations). (2) $\zeta(3)/\pi^3$ is not algebraic of degree ≤ 100 with coefficient norm $\leq 10^{183}$ (LLL), and specifically not of degree ≤ 30 with coefficient norm $\leq 10^{640}$. (3) ζ(3) and π satisfy no joint polynomial of total degree ≤ 12 with coefficient norm $\leq 10^{206}$, and specifically not of total degree ≤ 6 with coefficient norm $\leq 10^{710}$ (LLL). (4) No linear relation connects ζ(3), ζ(3,2), ζ(2,3), and π^5 at weight 5. (5) A broader search across five previously-untested constant families - including $\mathrm{Li}_4(1/2)$, whose closed form (if any) is a known open question - finds no relations, at LLL-certified bounds of 10^{398} to 10^{999} (§3.9f). (6) The same algebraicity/bivariate treatment given to ζ(3), extended for the first time to Catalan's constant G, to ζ(5) and ζ(7) individually, and to the Euler-Mascheroni constant γ (whose irrationality is itself unproven), likewise finds no relations (§3.9g). (7) We numerically verify the conjectured bases of the multiple zeta value algebra at every weight 0-10, computing the genuine depth-2 generators ζ(6,2) and ζ(8,2) to 1100 digits via a validated trigamma reduction. The weight-8 result targets the open lower-bound direction of Zagier's dimension conjecture: ζ(6,2) is not a rational combination of π^8, $\zeta(3)^2\pi^2$, $\zeta(3)\zeta(5)$ with coefficient norm below 10^{237}, so the weight-8 space really does look 4-dimensional as

Zagier predicts (dim ≤ 4 being already a theorem of Terasoma and Deligne-Goncharov). Specialized MZV numerics has verified this structure to far higher weight (the Data Mine of Blümlein-Broadhurst-Vermaseren reached weight 22); ours is an independent reproduction within this paper's certified-bound framework, not a frontier advance (§3.9h). All 37 PSLQ tests plus the LLL cross-validation, extended-search, new-constant, and Zagier-dimension suites are reported with full parameters; the entire LLL work runs in about 12 minutes.

Correction history (kept for transparency). An earlier version of this manuscript claimed the degree-25 (10^{200}), degree-30 (10^{100}), and bivariate degree-6 (10^{50}) bounds from PSLQ runs that never actually reached those bounds - the pitfall described above. Honest PSLQ re-verification (8.7-12.3 hours per test) produced only trivial bounds (17, 2, and 0 respectively), and those claims were withdrawn. The LLL results above re-establish all three exclusions with far stronger, genuinely certified bounds; the withdrawal history is preserved in §3.9.

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1. Introduction

The Riemann zeta function at s = 3,

$\zeta(3) = \Sigma(n=1 \text{ to } \infty)\ 1/n^3 = 1.202056903159594285399738161\,5...$

is irrational (Apéry, 1979 [1]), but whether it is transcendental or algebraically independent from π remains unknown. For even zeta values, Euler showed $\zeta(2k) \in \pi^{2k} \cdot \mathbb{Q}$. The simplest open case is whether ζ(3) bears any rational relationship to π^2 - that is, whether integers a, b, c exist with $a\cdot\zeta(3) + b\cdot\pi^2 + c = 0$. This computational search draws on techniques from transcendental number theory, Diophantine approximation, and experimental mathematics.

We address this question computationally. Our main result is:

> **No integers a, b, c with $|a|, |b|, |c| \leq 10^{2000}$ satisfy $a\cdot\zeta(3) + b\cdot\pi^2 + c = 0$.**

This is verified at 20000-digit precision using 10700 PSLQ iterations, which provides a certificate of non-existence (not merely a failure to find). An extended run at 40000 digits and 100000 iterations pushes this to $|a|, |b|, |c| \leq 10^{18695}$. Both bounds far exceed the coefficients appearing in any known zeta identity - for comparison, $\zeta(2) = \pi^2/6$ has coefficients 1 and 6.

We supplement this with tests against other odd zeta values, Nesterenko's algebraically independent triple {π, e^π, Γ(1/4)}, and the Euler sum constants $\pi^2\cdot\ln 2$ and $\ln^3 2$. All return null results within the tested bounds.

In the course of this project, an earlier draft of this manuscript fell into a reproducibility trap general enough to be worth reporting on its own terms: three tests reported PSLQ "certificates" of enormous bounds (10^{50}-10^{200}) that had never actually been reached, because raising the requested bound

(`maxcoeff`) without also raising the iteration budget (`maxsteps` , which defaults to 100 in mpmath) leaves the function returning `None` regardless of whether it certified anything or simply ran out of iterations - the two cases are indistinguishable without checking the algorithm's internal state. We formalize this as a general diagnostic (§2.1c), quantify how sharply it degrades with basis size, and demonstrate the fix (independent cross-validation via LLL, §2.1b). We consider this finding, and not any particular bound on ζ(3), the result most likely to be useful to readers who never intend to think about ζ(3) again.

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2. Methods

2.1 The PSLQ Algorithm

Given real numbers $x_1, \ldots, x_?$ computed to D decimal digits, the PSLQ algorithm [2] either:

(a) Finds integers $a_1, \ldots, a_?$ (not all zero) with $a_1x_1 + \ldots + a_?x_? = 0$, or
(b) **Certifies** that no such relation exists with $\max|a_i| \le M$.

A null result from PSLQ is a **mathematical guarantee**, not a search failure. This is the key distinction from heuristic methods. We verify this programmatically: for the main test ({ζ(3), π², 1} at 20000 digits, 10700 iterations), the algorithm's internal norm bound exceeds 10^{2000} before termination, confirming it exited via the norm-exceeds-maxcoeff condition rather than exhausting its iteration limit. This certifies that any integer relation must have max|coefficient| > 10^{2000}. Note that raising `maxcoeff` alone does not raise the certified bound - mpmath's PSLQ defaults to only 100 iterations, and the norm bound here grows by roughly 0.19 decimal digits per iteration, so reaching 10^{2000} required explicitly requesting 10700 iterations.

2.1b LLL Reduction (second, independent engine)

We additionally implement integer relation detection via LLL lattice reduction using fpylll (the compiled fplll library, as used by SageMath). Given $x_1, \ldots, x_?$ at D digits, we reduce the n × (n+1) lattice $[I_? \mid \mathrm{round}(N \cdot x_i)]$ with N = 10^S, S < D. Any integer relation makes the corresponding lattice vector short (~‖a‖ rather than the generic ~10^(S/n)), so LLL finds it; conversely, fplll's guarantee that its output is LLL-reduced yields a certified exclusion: any relation must satisfy $\|a\|_2 \ge \|b_1\|$ / (2^((n-1)/2)·√(1+n/4)), where b_1 is the shortest reduced vector. Note this bounds the Euclidean norm of the coefficient vector; the corresponding bound on $\max|a_i|$ is smaller by at most √n (under one order of magnitude for every basis in this paper).

One implementation caution, documented in `lll_tests.py` : when no relation exists, LLL returns "balanced" vectors whose residual $\Sigma a_i x_i \approx$ |last coordinate|/N is small automatically and does *not* indicate a relation. A genuine relation is distinguished by its residual being zero to the full precision of the inputs ($\leq \|a\|_1 \cdot 10^{-D}$); every candidate is verified against that threshold, with inputs computed 1000 digits above the scaling S.

The LLL engine is dramatically faster than mpmath's PSLQ on this workload - the entire cross-validation suite (`lll_tests.py`) runs in under 30 seconds, versus 30+ hours for the equivalent PSLQ re-verification runs - and, being an independent algorithm in an independent implementation, it cross-validates every PSLQ result it reproduces.

2.1c A General Diagnostic: The PSLQ Certification Pitfall

We believe the most broadly useful finding of this project is not about ζ(3) at all, and is worth stating as a general warning independent of any specific result in this paper.

The pitfall. mpmath's `pslq(basis, maxcoeff=M)` returns `None` in exactly two circumstances: (a) the algorithm's internal norm bound provably exceeded `M`, which is a rigorous certificate that no relation exists with coefficients below `M`; or (b) the algorithm exhausted its iteration budget (`maxsteps`, defaulting to 100) before the norm bound reached `M`, in which case `None` means nothing more than "not found in 100 iterations" - the same as a plain, uncertified failure. **These two outcomes are indistinguishable from the return value alone.** An earlier version of this manuscript fell into exactly this trap: three tests (§3.9a-b) reported bounds of 10^{50}-10^{200} that were never actually reached, because `maxsteps` was left at its default while `maxcoeff` was raised to an astronomical value. The function returned `None` every time, and every time it was silently wrong to call that a certificate.

Severity depends sharply on basis size, in a way that is easy to underestimate. The norm bound grows at a roughly constant rate *per iteration* for a fixed basis, but that rate collapses as the basis grows - the same finite precision is divided across more dimensions:

Basis size (n)	Measured rate	100 default iterations reach
3 (main test)	~0.19 digits/iteration	$\sim 10^{19}$
26 (degree-25 algebraicity)	~0.0004 digits/iteration	$\sim 10^{0.04}$ (i.e. nothing)
28 (bivariate degree-6)	0 (norm stayed ≤ 1 for 3000 iterations)	nothing
31 (degree-30 algebraicity)	~0.0001 digits/iteration	$\sim 10^{0.01}$ (i.e. nothing)

A basis of 26-31 elements is not exotic - the original manuscript's own degree-30 test used exactly this size - and yet the default settings certify essentially nothing on it. mpmath's documentation contains one oblique acknowledgment, in a worked example rather than a warning attached to the `pslq` signature itself: "It may also be necessary to set maxsteps high to prevent a premature exit (before the coefficient bound has been reached)." This correctly implies premature exit is possible, but does not explain that the resulting `None` is then silently indistinguishable from a genuine certificate, nor that the failure rate is basis-size-dependent in the way §2.1c quantifies. We checked mpmath's issue tracker (open and closed issues mentioning `pslq` or `maxsteps`) and found no prior report of this specific ambiguity, and have since filed one: mpmath/mpmath#1130.

This is a real gap in mpmath's implementation specifically, not an unsolved problem in PSLQ generally. The theoretical distinction between "norm bound exceeded `maxcoeff`" (rigorous) and "iteration budget exhausted" (not rigorous) is well understood in the PSLQ literature (e.g. the original algorithm description [2] discusses both termination conditions), and at least one other PSLQ implementation we are aware of (the Go package `github.com/ncw/pslq`) returns a distinct `ErrorIterationsExceeded` rather than conflating the two outcomes into a single ambiguous result. mpmath's `pslq` does not make this distinction available without manually parsing `verbose=True` output and comparing the printed iteration count to `maxsteps` - which is what we now do.

The diagnostic. Before treating any `pslq(...) is None` result as a certificate, check the algorithm's actual termination condition (mpmath exposes this via `verbose=True`; we implement this check in `run_tests.py` §9): if the norm bound reached `maxcoeff`, the result is rigorous; if `maxsteps` was exhausted first, it is not, regardless of what `maxcoeff` was requested.

The fix we demonstrate. LLL reduction (§2.1b) does not share this failure mode in the same form - a completed LLL reduction always yields a genuine, computable exclusion bound from the reduced basis, even if that bound is smaller than hoped. Where PSLQ needed 8.7-12.3 hours to produce a near-worthless bound on these bases, LLL produced a bound exceeding the original (never-achieved) claim in single-digit seconds (§3.9). We recommend, as general practice: (1) always verify PSLQ's termination mode rather than trusting a `None` return; (2) treat iteration-budget requirements as scaling badly, not linearly, with basis size; (3) cross-validate any large-basis null result with an independent method before publishing it as a certificate.

2.2 Computational Setup

- **Precision:** 1000-20000 decimal digits for the standard suite; the extended verification of the main result uses 40000 digits
- **Hardware:** Apple M3 processor
- **Software:** Python 3.14, mpmath 1.4.1, fpylll 0.6.4 (with gmpy2 2.3.1). The PSLQ results were computed on mpmath's default pure-Python big-integer backend; `gmpy2` is auto-detected by mpmath and gives a benchmarked 28.1x speedup with no code changes, and is required for the LLL suite's high-precision constant computation to be fast
- **PSLQ iterations:** mpmath's `pslq` defaults to 100 iterations, sufficient only for tests with modest bounds ($\lesssim 10^{12}$). Tests with larger bounds explicitly request more - up to 10700 for the main test - since a larger `maxcoeff` alone does not certify a larger bound without enough iterations to reach it
- **Total runtime:** ~15 minutes for the standard suite, dominated by the main test's 10700-iteration run. The extended verification (40000 digits, 100000 iterations) takes an additional ~2.9 hours and is not part of the standard suite

2.3 Validation

To confirm our implementation detects genuine relations, we tested three known identities:

1. **$\mathrm{Li}_3(1/2)$:** PSLQ recovered [21, −24, −2, 4] for the basis $\{\zeta(3), \mathrm{Li}_3(1/2), \pi^2\ln 2, \ln^3 2\}$, with residual 6.4×10^{-401}. This corresponds to $\zeta(3) = (8/7)\mathrm{Li}_3(1/2) + (2/21)\pi^2\ln 2 - (4/21)\ln^3 2$.

2. **$\mathrm{Li}_3(-1)$:** PSLQ recovered [3, 4] for the basis $\{\zeta(3), \mathrm{Li}_3(-1)\}$, confirming $\mathrm{Li}_3(-1) = -3\zeta(3)/4$.

3. **$\zeta(6)$:** PSLQ recovered [0, −945, 1, 0] for the basis $\{\zeta(3)^2, \zeta(6), \pi^6, 1\}$, confirming $\zeta(6) = \pi^6/945$.

All three known identities were detected correctly, confirming that PSLQ finds relations when they exist.

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3. Results

3.1 Main Results

The strongest result of this paper:

> **Result A. At 20000-digit precision, using 10700 PSLQ iterations (mpmath's default of 100 is far too few to reach this bound), no relation $a\cdot\zeta(3) + b\cdot\pi^2 + c = 0$ exists with $|a|, |b|, |c| \le 10^{2000}$. The PSLQ norm bound certifies non-existence. This takes approximately 7.6 minutes on an Apple M3.**
>
> **Result A (extended). Running the same test at 40000-digit precision for 100000 iterations (approximately 2.9 hours on an Apple M3) extends this to $|a|, |b|, |c| \le 10^{18695}$. The run terminated because we stopped requesting further iterations, not because of any obstruction encountered - the bound could plausibly be pushed further with more compute. We report it as a secondary, more expensive verification rather than the paper's primary reproducible claim.**
>
> **Result A (LLL cross-validation). The independent LLL engine certifies $\|a\|_2 \ge 10^{19999}$ for the same basis (scale 60000 digits, 0.23s of reduction time after 3.7s of constant computation). Two independent algorithms in independent implementations thus agree: no relation $a\cdot\zeta(3) + b\cdot\pi^2 + c = 0$ exists with coefficients below at least 10^{18695}.**
>
> **Result B (re-established via LLL). $\zeta(3)/\pi^3$ is not algebraic of degree ≤ 30 with coefficient norm $\|a\|_2 \le 10^{640}$ (LLL, scale 20000 digits, 7.6s), nor of degree ≤ 25 with $\|a\|_2 \le 10^{765}$ (4.4s). *History:* an earlier version claimed degree ≤ 30 at height 10^{100} from a PSLQ run that never reached that bound; honest PSLQ re-verification produced only a trivial bound of 2 after 12.3 hours (the 31-element basis dilutes mpmath's PSLQ beyond**

> **practical use - §3.9a), and the claim was withdrawn before being re-established, far more strongly, by the LLL engine.**

For context: $\zeta(2)/\pi^2 = 1/6$ is rational (degree 0). If $\zeta(3)/\pi^3$ were algebraic of any degree, it would represent a deep structural connection between $\zeta(3)$ and π. We now exclude this up to degree 30 with genuinely certified bounds.

3.2 Complete List of Tested Bases

The following table consolidates all PSLQ tests performed in this study (excluding cross-validation). Tests 32-34 recover known identities; all others certify non-existence.

#	Basis	Size	Digits	Bound	Result
1	$\{\zeta(3), \pi^2, 1\}$	3	20000	10^{2000}	No relation
1'	$\{\zeta(3), \pi^2, 1\}$ (extended)	3	40000	10^{18695}	No relation
2	$\{\zeta(3), \pi^3, 1\}$	3	5000	10^{1000}	No relation
3	$\{\zeta(3), \pi^2, \pi^4, 1\}$	4	2000	10^{10}	No relation
4	$\{\zeta(3), \pi^2, \pi^4, \pi^6, 1\}$	5	1000	10^8	No relation
5	$\{\zeta(3), \pi^2, \pi^4, \pi^6, \pi^8, \pi^{10}, 1\}$	7	3000	10^8	No relation
6	$\{1, \zeta(3), \zeta(3)^2\}$	3	2000	10^{12}	No relation
7	$\{1, \zeta(3), \zeta(3)^2, \zeta(3)^3\}$	4	2000	10^{10}	No relation
8	$\{1, \zeta(3), \zeta(3)^2, \zeta(3)^3, \zeta(3)^4\}$	5	2000	10^8	No relation
9	$\{(\zeta(3)/\pi^3)^k : k=0..10\}$	11	8000	10^{12}	No relation
10	$\{(\zeta(3)/\pi^3)^k : k=0..15\}$	16	10000	10^9	No relation
11	$\{(\zeta(3)/\pi^3)^k : k=0..25\}$	26	50000	17 (PSLQ)	No relation; LLL certifies 10^{765} (§3.9)
12	$\{(\zeta(3)/\pi^3)^k : k=0..30\}$	31	50000	2 (PSLQ)	No relation; LLL certifies 10^{640} (§3.9)
13	$\{\zeta(3)^i\pi^j : i+j\leq3\}$	10	5000	10^{12}	No relation; LLL: 10^{1998}
14	$\{\zeta(3)^i\pi^j : i+j\leq4\}$	15	5000	10^8	No relation; LLL: 10^{1331}
15	$\{\zeta(3)^i\pi^j : i+j\leq6\}$	28	50000	0 (PSLQ)	No relation; LLL certifies 10^{710} (§3.9)
16	$\{\zeta(3), \zeta(5), 1\}$	3	3000	10^{12}	No relation
17	$\{\zeta(3), \zeta(5), \zeta(7), 1\}$	4	3000	10^{10}	No relation
18	$\{\zeta(3), \zeta(5), \zeta(7), \zeta(9), 1\}$	5	1500	10^8	No relation
19	$\{\zeta(3), \pi, e^\pi, \Gamma(1/4), 1\}$	5	4000	10^{12}	No relation
20	$\{\zeta(3), \Gamma(1/4)^4/\pi^3, \pi^2, 1\}$	4	4000	10^{12}	No relation

#	Basis	Size	Digits	Bound	Result
21	$\{\zeta(3), G, \pi^3, 1\}$	4	3000	10^{10}	No relation
22	$\{\zeta(3), G^2, G\cdot\pi, \pi^2, G, 1\}$	6	3000	10^{10}	No relation
23	$\{\zeta(3)^2, G^2, \zeta(3)\cdot G, \pi^4, \zeta(3), G, \pi^2, 1\}$	8	2000	10^8	No relation
24	$\{\zeta(3)^2, \zeta(5)\cdot\pi, \zeta(7), \pi^6, \pi^4, 1\}$	6	3000	10^8	No relation
25	$\{\zeta(3), \zeta(3)\cdot\pi^2, \zeta(5)\cdot\pi^2, \zeta(5), \pi^4, \pi^2, 1\}$	7	3000	10^8	No relation
26	$\{\zeta(3)^2, \zeta(5), \pi^6, \pi^4, \pi^2, 1\}$	6	3000	10^{10}	No relation
27	$\{\zeta(3), \zeta(2)\zeta(3)-\zeta(5), \zeta(5), \pi^2, 1\}$	5	1000	10^8	No relation
28	$\{\zeta(3), L(E_{32},2), \pi^2, 1\}$	4	2000	10^{10}	No relation
29	$\{\zeta(3), L(\chi_{-4},3), \pi^2, 1\}$	4	2000	10^{10}	No relation
30	$\{\zeta(3), L(E_{32},2), L(\chi_{-4},3), \pi^2, 1\}$	5	2000	10^8	No relation
31	$\{\zeta(3), \pi^2\ln 2, \ln^3 2, \ln^2 2, \ln 2, \pi^2, 1\}$	7	4000	10^{11}	No relation
32	$\{\zeta(3), \mathrm{Li}_3(1/3), \pi^2\ln 3, \ln^3 3, 1\}$	5	2000	10^{10}	No relation
33	$\{\zeta(3), \mathrm{Li}_3(1/4), \pi^2\ln 2, \ln^3 2, 1\}$	5	2000	10^{10}	No relation
34	$\{\zeta(3), \zeta(3,2), \zeta(2,3), \pi^5, 1\}$	5	4000	10^{10}	No relation
35	$\{\zeta(3), \mathrm{Li}_3(1/2), \pi^2\ln 2, \ln^3 2\}$	4	5000	10^6	**FOUND** [21,-24,-2,4]
36	$\{\zeta(3), \mathrm{Li}_3(-1)\}$	2	5000	10^6	**FOUND** [3, 4]
37	$\{\zeta(3)^2, \zeta(6), \pi^6, 1\}$	4	5000	10^6	**FOUND** [0,-945,1,0]

3.3 Linear Independence from π

Result 3.3. *No relation $a\cdot\zeta(3) + b\cdot\pi^2 + c = 0$ exists with $|a|, |b|, |c| \leq 10^{2000}$ (20000 digits, 10700 iterations; extended to 10^{18695} at 40000 digits and 100000 iterations). No relation $a\cdot\zeta(3) + b\cdot\pi^3 + c = 0$ exists with $|\text{coefficients}| \leq 10^{1000}$ (5000 digits).*

Basis	Bound	Precision
$\{\zeta(3), \pi^2, 1\}$	10^{2000}	20000
$\{\zeta(3), \pi^2, 1\}$ (extended)	10^{18695}	40000
$\{\zeta(3), \pi^3, 1\}$	10^{1000}	5000
$\{\zeta(3), \pi^2, \pi^4, 1\}$	10^{10}	2000
$\{\zeta(3), \pi^2, \pi^4, \pi^6, 1\}$	10^8	1000
$\{\zeta(3), \pi^2, \pi^4, \pi^6, \pi^8, \pi^{10}, 1\}$	10^8	3000

3.4 Algebraicity of ζ(3) and ζ(3)/π³

Result 3.4a. *$\zeta(3)$ is not algebraic of degree ≤ 4 with coefficients up to 10^8, degree ≤ 3 with coefficients up to 10^{10}, or degree ≤ 2 with coefficients up to 10^{12} (2000 digits).*

Result 3.4b. *$\zeta(3)/\pi^3$ is not algebraic of degree ≤ 10 with height $\leq 10^{12}$ (PSLQ, 8000 digits; LLL cross-validation strengthens this to $\|a\|_2 \leq 10^{1816}$), or degree ≤ 15 with height $\leq 10^9$ (PSLQ, 10000 digits; LLL: $\|a\|_2 \leq 10^{1247}$). Via LLL, additionally not of degree ≤ 25 with $\|a\|_2 \leq 10^{765}$ or degree ≤ 30 with $\|a\|_2 \leq 10^{640}$ (see §3.9a for the history of these last two).*

All four degree tests are now certified by the LLL engine (§2.1b), which is unaffected by the large-basis dilution that limited mpmath's PSLQ; the degree ≤ 10 and degree ≤ 15 tests are additionally confirmed by PSLQ, giving two-engine cross-validation at those degrees.

3.5 Bivariate Polynomial Independence

Result 3.5. *$\zeta(3)$ and π satisfy no joint polynomial equation for the following parameters:*

Total degree	Basis size	PSLQ bound	LLL bound ($\|a\|_2$)
≤ 3	10	10^{12} (5000 digits)	10^{1998}
≤ 4	15	10^8 (5000 digits)	10^{1331}
≤ 6	28	- (see §3.9b history)	10^{710}

The degree ≤ 6 exclusion comes from the LLL engine only: the original PSLQ claim (10^{50}) was withdrawn after honest re-verification produced a certified bound of exactly 0 (§3.9b), and mpmath's PSLQ cannot certify anything useful for this 28-element basis in practical time. The degree ≤ 3 and ≤ 4 rows are confirmed by both engines.

These tests directly address whether ζ(3) and π are algebraically dependent.

3.6 Multiple Zeta Values at Weight 5

Multiple zeta values (MZVs) are nested sums ζ(s_1, s_2, ...) = Σ_{m_1>m_2>...≥1} 1/(m_1^s_1 m_2^s_2 ...). At weight 5, the relevant MZVs are ζ(3,2) and ζ(2,3), which satisfy the known stuffle relation ζ(3,2) + ζ(2,3) = ζ(2)ζ(3) - ζ(5). We test whether ζ(3) satisfies any additional linear relation with these values and π^5.

Result 3.6. *At 4000-digit precision, no relation*

$$a\cdot\zeta(3) + b\cdot\zeta(3,2) + c\cdot\zeta(2,3) + d\cdot\pi^5 + e = 0$$

exists with $|a|, |b|, |c|, |d|, |e| \leq 10^{10}$ (0.74s).

This is an additional data point. Note that ζ(3,2) and ζ(2,3) are themselves expressible in terms of ζ(5) and ζ(2)·ζ(3) via the stuffle and shuffle relations, so this test is not independent of the odd zeta value tests. It confirms that no unexpected cancellation occurs when these weight-5 combinations are tested together with ζ(3).

3.7 Independence from Other Constants

Result 3.6a (Odd zeta values). *No linear relation connects:*

Basis	Bound	Precision
{ζ(3), ζ(5), 1}	10^{12}	3000
{ζ(3), ζ(5), ζ(7), 1}	10^{10}	3000
{ζ(3), ζ(5), ζ(7), ζ(9), 1}	10^{8}	1500

Result 3.6b (Nesterenko's triple). *No relation a·ζ(3) + b·π + c·e^π + d·Γ(1/4) + f = 0 exists with |coefficients| ≤ 10^{12} (4000 digits).*

Result 3.6c (Catalan's constant). *No relation a·ζ(3) + b·G + c·π^3 + d = 0 exists with |coefficients| ≤ 10^{10} (3000 digits). No quadratic relation involving G^2, G·π, π^2, G exists with the same bounds.*

Result 3.6d (Lemniscate constant). *No relation a·ζ(3) + b·$\Gamma(1/4)^4/\pi^3$ + c·π^2 + d = 0 exists with |coefficients| ≤ 10^{12} (4000 digits).*

Result 3.6e (Product relations). *No relation connects $\zeta(3)^2$ to ζ(5)·π or ζ(7) modulo π^6, π^4 with |coefficients| ≤ 10^8 (3000 digits). No depth-graded relation ζ(3)·(1+a·π^2) = b·ζ(5)·π^2 + ... exists with the same bounds.*

3.8 L-Values of Elliptic Curves

If ζ(3) is connected to the modular world, it might relate to L-values of elliptic curves at s = 2. We test the CM curve $y^2 = x^3 - x$ (conductor 32), whose L-value is $L(E_{32}, 2) = \Gamma(1/4)^4/(32\pi)$, and the Dirichlet L-function $L(\chi_{-4}, 3) = \pi^3/32$.

Result 3.7. *At 2000-digit precision, no relation connects ζ(3) to: - $L(E_{32}, 2)$ and π^2 with |coefficients| ≤ 10^{10} - $L(\chi_{-4}, 3)$ and π^2 with |coefficients| ≤ 10^{10} - Both L-values simultaneously with |coefficients| ≤ 10^8*

ζ(3) is not a rational linear combination of these L-values and π^2.

3.9 Higher-Degree and Harder Tests

Result 3.9a (revised - inconclusive). *The original manuscript claimed ζ(3)/π^3 is not algebraic of degree ≤ 25 with polynomial height ≤ 10^{200} at 6000 digits in 27.0s. This was never actually achieved:*

mpmath's PSLQ defaults to 100 iterations, and reaching a norm bound of 10^{200} was never run to completion. We re-ran this test properly: at 50000-digit precision using the full 3000-iteration budget we allotted (8.7 hours of compute), the certified norm bound reached only 17 - i.e., we can only certify the absence of a degree-25 relation with |coefficients| ≤ 17, not 10^{200}. This is a consequence of the 26-element basis: precision is divided across far more dimensions than in the main test, and the norm grows roughly 280× slower per iteration (about 0.0007 decimal digits/iteration here, versus 0.19 for the 3-element main-test basis). Reaching a bound like 10^8 at this rate would require on the order of 10^5 iterations - roughly 12 days of compute at the same per-iteration cost - which we consider impractical for this paper. We report this test as inconclusive rather than as a meaningful exclusion result, and flag it as an open item for future work (e.g. a compiled LLL implementation such as fpylll may scale better on large bases than mpmath's fixed-point PSLQ).

Result 3.9a' (degree-30: withdrawn). *The original manuscript's headline "Result B" claimed $\zeta(3)/\pi^3$ is not algebraic of degree ≤ 30 with polynomial height ≤ 10^{100} at 4500 digits in 29.4s - never actually achieved for the same reason as above. We re-ran it at 50000-digit precision with a full 3000-iteration budget (12.3 hours of compute): the certified norm bound reached only 2. This is the largest basis in the paper (31 elements) and shows the worst dilution of the three affected tests - worse even than the degree-25 test above. We withdraw this claim entirely; it was one of the paper's two headline results and should not be cited.*

Result 3.9b (withdrawn). *The original manuscript claimed that at 4000-digit precision, ζ(3) and π satisfy no joint polynomial of total degree ≤ 6 with |coefficients| ≤ 10^{50} (28-element basis, 18.1s). This was never actually achieved - mpmath's PSLQ defaults to 100 iterations, and a run reaching norm 10^{50} was never completed. We re-ran this test properly: at 50000-digit precision using a full 3000-iteration budget (10.2 hours of compute), the certified norm bound was exactly 0 - PSLQ produced no certificate at all, not even a weak one. This is the same large-basis dilution problem as Result 3.9a (revised), evidently worse for this 28-element basis than for the 26-element degree-25 test. We withdraw this claim; it should not be cited.*

Result 3.9 (final): all three exclusions re-established via LLL. The withdrawn/inconclusive entries above (3.9a, 3.9a', 3.9b) record the honest status of these tests under mpmath's PSLQ, and we preserve them for transparency. They are now superseded: the LLL engine (§2.1b, `lll_tests.py`) certifies all three exclusions in seconds, at bounds far beyond both the original fabricated claims and anything PSLQ achieved in 30+ hours of re-verification:

Test	Original claim (never achieved)	Honest PSLQ bound (hours)	LLL certified bound (seconds)
Degree-25 algebraicity	10^{200}	17 (8.7h)	$\|\mathbf{a}\|_2 \geq 10^{765}$ (4.4s)
Degree-30 algebraicity	10^{100}	2 (12.3h)	$\|\mathbf{a}\|_2 \geq 10^{640}$ (7.6s)

Test	Original claim (never achieved)	Honest PSLQ bound (hours)	LLL certified bound (seconds)
Bivariate degree-6	10^{50}	0 (10.2h)	**$\|a\|_2 \geq 10^{710}$** (5.5s)

All LLL runs use scale S = 20000 digits with constants computed at 21000 digits; the certificate is the LLL-reducedness guarantee of fplll plus the proven approximation factor (§2.1b). An earlier interim note here proposed re-attempting these tests with the 28x-faster gmpy2 PSLQ backend; the LLL approach made that unnecessary.

Result 3.9c (Weight 6). *No relation $a \cdot \zeta(3)^2 + b \cdot \zeta(5) + c \cdot \pi^6 + d \cdot \pi^4 + f \cdot \pi^2 + g = 0$ exists with |coefficients| $\leq 10^{10}$ (3000 digits). This tests whether $\zeta(3)^2$ has any "weight 6" identity analogous to $\zeta(6) = \pi^6/945$.*

Result 3.9d (Multiple zeta values). *No relation connects $\zeta(3)$ to $\zeta(2) \cdot \zeta(3) - \zeta(5)$ (the sum $\zeta(3,2) + \zeta(2,3)$) with |coefficients| $\leq 10^8$ (1000 digits).*

Result 3.9e (Catalan quadratic). *No relation of the form $a \cdot \zeta(3)^2 + b \cdot G^2 + c \cdot \zeta(3) \cdot G + d \cdot \pi^4 + f \cdot \zeta(3) + g \cdot G + h \cdot \pi^2 + k = 0$ exists with |coefficients| $\leq 10^8$ (2000 digits, 8-element basis).*

3.9f Extended Verification: Higher Degree and a Broader Search

Section 3.9's LLL results made it practical to ask two further questions cheaply (whole sweep: 10.2 minutes, `lll_extended.py`): (1) does the degree-30 and degree-6 boundary reflect a real obstruction, or just where the original PSLQ-based paper happened to stop? (2) can the same LLL machinery search genuinely new territory rather than re-confirming the four relation families already tested?

Result 3.9f-i (higher degree). *At the same scale (S = 20000 digits) used for the degree-25/30 tests above, we pushed further:*

Test	Basis size	Original stopping point	Extended result
$\zeta(3)/\pi^3$ algebraicity	101 (degree $\leq$ 100)	degree $\leq$ 30, $\|a\|_2 \geq 10^{640}$	**degree $\leq$ 100, $\|a\|_2 \geq 10^{183.1}$** (355s)
Bivariate $\zeta(3)^i \pi^j$	91 (total degree $\leq$ 12)	degree $\leq$ 6, $\|a\|_2 \geq 10^{710}$	**degree $\leq$ 12, $\|a\|_2 \geq 10^{206.4}$** (250s)

The exclusion bound shrinks as degree grows (fixed precision is divided across more basis dimensions - the same phenomenon that broke mpmath's PSLQ, just far less severely for LLL), but even at degree 100 the bound (10^{183}) still exceeds the original manuscript's degree-30 claim (10^{100}) while excluding more than 3× the degree. The original degree $\leq$ 30 / degree $\leq$ 6 stopping points reflected the 2005-era PSLQ literature's practical limits, not a mathematical obstruction.

Result 3.9f-ii (broader search: new constant families). *We extended the search to five families not tested elsewhere in this paper, all confirmed algebraically independent (or, for $Li_4(1/2)$, unrelated to the tested basis) within LLL-certified bounds of 10^{398} to 10^{999}:*

Test	Basis	Bound
$Li_4(1/2)$ vs $\{\pi^4, \ln^4 2, \pi^2\ln^2 2, \zeta(3)\ln 2\}$	5	10^{799}
$Li_3(1/5)$, $Li_3(1/6)$, $Li_3(1/7)$, $Li_3(2/3)$, $Li_3(3/4)$ vs $\{\zeta(3), \pi^2\ln(x), \ln^3(x), 1\}$	5 each	10^{799}
$\zeta(3)$ vs $L(\chi_{-3},2)$, $L(\chi_{-3},3)$, π^2	5	10^{799}
$\zeta(3)$ vs $\Gamma(1/3)^6/\pi^4$, π^2	4	10^{999}
$\zeta(3)$ vs $\Gamma(1/5)$, $\Gamma(2/5)$, π, $\sqrt{5}$	6	10^{665}
Kitchen sink: $\{\zeta(3), \zeta(5), \zeta(7), \pi, \ln 2, G, Li_3(1/2), e^\pi, \Gamma(1/4), 1\}$	10	10^{398}

$Li_4(1/2)$ is of particular interest: unlike $Li_2(1/2) = \pi^2/12 - \ln^2(2)/2$ and $Li_3(1/2)$ (Result, §2.3), no closed form for $Li_4(1/2)$ in terms of $\zeta(4)$, π, and $\ln(2)$ is known in the literature. Our null result adds computational weight to that open question rather than resolving it - a negative result consistent with, not proof of, its conjectured irreducibility. The new Dirichlet L-values $L(\chi_{-3},s)$ are computed via the standard Hurwitz zeta decomposition $L(\chi,s) = k^{-s}\cdot\Sigma_r\chi(r)\zeta(s,r/k)$; we validated this method in-session against this paper's own established $L(\chi_{-4},3) = \pi^3/32$ (Result 3.7), matching to 40+ digits before using it on the untested character.

3.9g New Constant Families: Catalan, Higher Odd Zetas, and γ

Sections 3.5-3.9 give $\zeta(3)$ the full algebraicity-and-bivariate treatment against π, but three natural extensions of the same method to other constants had not been tried: does Catalan's constant G, tested elsewhere in this paper only in linear/quadratic combinations with $\zeta(3)$ (§3.7), satisfy an algebraicity-with-π or bivariate-with-π relation the way $\zeta(3)$ does? Do $\zeta(5)$ and $\zeta(7)$ - tested only for *linear* independence from π and each other (§3.7) - satisfy the same kind of nonlinear (algebraicity or joint-polynomial) relation? And what about the Euler-Mascheroni constant γ, which appears nowhere else in this paper and whose irrationality is not even known (unlike $\zeta(3)$, proved irrational by Apéry)? `lll_new_constants.py` tests all three (70.7s total):

Result 3.9g-i (Catalan's constant).

Test	Basis size	Bound
G/π^2 algebraic, degree ≤ 30	31	$10^{640.3}$ (7.4s)
Bivariate $G^i\pi^j$, total degree ≤ 8	45	$10^{437.7}$ (24.9s)

Result 3.9g-ii ($\zeta(5)$ and $\zeta(7)$ beyond linear independence).

Test	Basis size	Bound
$\zeta(5)/\pi^5$ algebraic, degree ≤ 30	31	$10^{640.4}$ (7.5s)
$\zeta(7)/\pi^7$ algebraic, degree ≤ 30	31	$10^{640.4}$ (7.4s)

Test	Basis size	Bound
Bivariate $\zeta(5)^i\zeta(7)^j$, total degree ≤ 5	21	$10^{187.2}$ (0.2s)

Result 3.9g-iii (Euler-Mascheroni constant γ).

Test	Basis size	Bound
$a\cdot\gamma + b\cdot\pi + c\cdot e + d = 0$ (linear)	4	$10^{999.4}$
γ algebraic, degree ≤ 5	6	$10^{665.6}$
Trivariate $\gamma^i\pi^j e^k$, total degree ≤ 6	84	$10^{35.2}$ (21.9s)

None of these three families is proved irrational, let alone transcendental, by these results - the same caveat as everywhere else in this paper (an exclusion within a bound is not a proof of independence). γ is the most notable: whether γ is even irrational is a famous open problem, so a null result here carries a different flavor of uncertainty than the ζ(3) tests - we cannot rule out that γ, π, and e are related by some mechanism these polynomial/algebraicity searches are structurally unable to detect (e.g. a relation depending on γ's still-unknown arithmetic nature). We report it as one more data point, not as evidence toward γ's irrationality.

3.9h Zagier's Dimension Conjecture: Verified Through Weight 10, Targeting the Open Direction at Weights 8 and 10

Every result so far in this paper is an exclusion: a relation was searched for and not found. This section reports something structurally different - a verification of the conjectured bases of the multiple zeta value algebra at every weight through 10, including, at weights 8 and 10, numerical evidence bearing on the *open* direction of Zagier's conjecture.

Background. Multiple zeta values (MZVs) generalize ζ(3) to nested sums of arbitrary depth; $\zeta(s_1,s_2) = \Sigma_{m>n\geq1} m^{-s_1} n^{-s_2}$ (we use the decreasing-index convention throughout this section; note the weight-5 test in §3.6 labels its values in the opposite convention, as parts of the literature do). Euler proved that every MZV of weight ≤ 7 is a polynomial in single zeta values. Zagier's dimension conjecture predicts the dimension d_w of the weight-w graded piece of the MZV algebra via $d_w = d_{w-2} + d_{w-3}$ ($d_0=1$, $d_1=0$, $d_2=1$): 1, 0, 1, 1, 1, 2, 2, 3, 4, 5, 7 for w = 0, ..., 10. Critically, the two directions of this conjecture have different statuses: **dim ≤ d_w is a theorem** (Terasoma; Deligne-Goncharov), while **dim ≥ d_w - that the conjectured generators are actually ℚ-linearly independent - is completely open** (even the irrationality of $\zeta(5)/\pi^5$ is unproven).

Weights 0-7 (a theorem, re-confirmed). The number of monomials in {π^2, ζ(3), ζ(5), ζ(7), ζ(9), ζ(11)} at weight w is a partition count that matches d_w exactly for w = 0-7 and w = 9, falling short at w = 8 (3 vs. 4) and w = 10 (5 vs. 7) - the classically known weights where a generator not expressible in single zetas first appears. LLL confirms the monomials are genuinely independent as real numbers:

Weight	Monomials	Zagier d_w	LLL bound
5	2	2	$10^{2000.1}$
6	2	2	$10^{2001.0}$
7	3	3	$10^{1333.4}$

Weights 8-10 (the genuine tests). The conventional extra generators are ζ(6,2) at weight 8 and ζ(8,2) at weight 10 - both **depth-2** MZVs (an earlier version of this manuscript wrongly claimed depth ≥ 3 was required and deferred these tests as infeasible; we record the correction here). Depth-2 values ζ(a,2) reduce, via the trigamma identity H $^{(2)}$ _{m−1} = ζ(2) − ψ′(m), to ζ(a,2) = ζ(2)ζ(a) − Σ_m ψ′(m)/m^a, whose tail is smooth and Euler-Maclaurin-summable to arbitrary precision with mpmath's `nsum` - no dedicated evaluator was needed after all. Before use, the evaluator was validated against three known closed forms (ζ(2,2) = π^4/120; ζ(4,2) = ζ(3)² − (4/3)ζ(6); ζ(3,2) = 3ζ(2)ζ(3) − (11/2)ζ(5)), each reproduced to 400+ digits, with LLL independently re-discovering the first two as identities; ζ(6,2) was additionally cross-checked against a brute-force truncated double sum to the expected truncation error.

Result 3.9h. *With ζ(6,2) and ζ(8,2) computed to 1100 digits and all other values at matching precision (LLL scale 950 digits), no rational relation exists within the certified bounds among the conjectured Zagier basis at each weight:*

Weight	Conjectured basis (d_w elements)	LLL bound
8	π^8, ζ(3)²π², ζ(3)ζ(5), **ζ(6,2)**	$10^{237.7}$
9	ζ(9), ζ(7)π², ζ(5)π^4, ζ(3)π^6, ζ(3)³	$10^{189.8}$
10	π^{10}, ζ(3)²π^4, ζ(3)ζ(5)π², ζ(3)ζ(7), ζ(5)², **ζ(6,2)π²**, **ζ(8,2)**	$10^{135.2}$

The weight-8 row is the one with real content: if ζ(6,2) were a rational combination of the three single-zeta monomials, the weight-8 MZV space would be 3-dimensional and Zagier's conjecture false. Our certificate excludes any such relation with coefficient norm below 10^{237}. Since dim ≤ 4 is already a theorem, this is numerical evidence for the open lower-bound direction: the weight-8 space really does look 4-dimensional. The same applies to ζ(8,2) at weight 10. These are the only results in this paper that bear directly on the open direction of a named conjecture, rather than excluding relations no one expected or re-confirming known theorems.

Scope relative to prior MZV numerics. This verification is new to this paper, not to the field. The Multiple Zeta Value Data Mine of Blümlein, Broadhurst, and Vermaseren [8] numerically verified the conjectured dimension structure of the MZV algebra to far higher weight (weight 22, and in restricted settings beyond), using dedicated high-performance evaluators much more capable than the trigamma reduction used here. Our weight-10 reach is nowhere near that frontier and does not claim to be. What this section contributes is (a) an independent, from-scratch reproduction inside this paper's certified-exclusion-bound framework, with the validation chain documented above, and (b) a demonstration that the LLL infrastructure built for the $\zeta(3)$-π search extends to structural conjecture-testing at essentially zero marginal cost. Readers interested in the state of the art in MZV verification should consult [8], not this section.

A cautionary note from this section's own development. Our first run of these tests produced plausible-looking "relations" with 25-43-digit coefficients at every weight. They were spurious: we had run LLL with only a 40-digit margin between the input precision and the lattice scale, and the spurious-vector filter in `lll_tests.py` needs a margin exceeding ~50 digits to reject balanced no-relation vectors (their coefficient sizes, 10^(S/n), matched the generic prediction exactly, which is what gave them away). The filter now raises an error on margins under 100 digits rather than silently mis-certifying. We mention this because it is the LLL-side analogue of the PSLQ pitfall of §2.1c: every certification method in this paper has at least one silent failure mode, and both were caught only by knowing what the failure signature looks like.

3.10 BBP-Type Formula Search

Result 3.10a (Validation). *PSLQ recovers the known identity [21, −24, −2, 4] for {$\zeta(3)$, $Li_3(1/2)$, $\pi^2\ln 2$, $\ln^3 2$} at 5000 digits.*

Result 3.10b. *Without $Li_3(1/2)$, no relation*

$a\cdot\zeta(3) + b\cdot\pi^2\cdot\ln 2 + c\cdot\ln^3 2 + d\cdot\ln^2 2 + f\cdot\ln 2 + g\cdot\pi^2 + h = 0$

exists with |coefficients| $\leq 10^{11}$ (4000 digits). $\zeta(3)$ has no BBP-type formula that avoids $Li_3(1/2)$.

Result 3.10c. *$Li_3(1/4)$ cannot substitute for $Li_3(1/2)$ - it receives coefficient 0 when both are in the basis. $Li_3(1/3)$ similarly has no identity connecting it to $\zeta(3)$ with |coefficients| $\leq 10^{10}$ (2000 digits).*

3.11 Continued Fraction Analysis

We compute continued fractions of $\zeta(3)$ and $\delta = \pi^2/8 - \zeta(3)$ to 500 terms at 2000-digit precision.

Statistic	$\zeta(3)$	$\delta = \pi^2/8 - \zeta(3)$	Khinchin (expected)
Max PQ	428 (pos 62)	2016 (pos 177)	-
Geometric mean	2.81	2.73	2.69
% equal to 1	42.2%	43.0%	41.5%

Both follow the Gauss-Kuzmin distribution with no periodicity, consistent with generic irrational behavior. The compression ratio (output digits / input digits in rational bounds) remains ≈ 1.0 at all scales - no exceptional approximation exists.

Selected rational bounds on $\zeta(3) = \pi^2/8 - \delta$:

Digits	Lower	Upper
19.8	$\pi^2/8 - 40545279/1281308663$	$\pi^2/8 - 1585749442/50112726993$
28.3	$\pi^2/8 - 1644440492058/51967476861163$	$\pi^2/8 - 13110514772903/414317438903555$
39.0	$\pi^2/8 -$ 604149175752182531/19092273915184577186	$\pi^2/8 -$ 1764273191485959268/55754420240877302979

3.12 Statistical Normality and Cross-Validation

Over 9,000 decimal digits, ζ(3) passes the chi-squared normality test ($\chi^2 = 11.23$, critical value 16.92). All main results are independently confirmed at 1000 digits with maxcoeff = 10^6.

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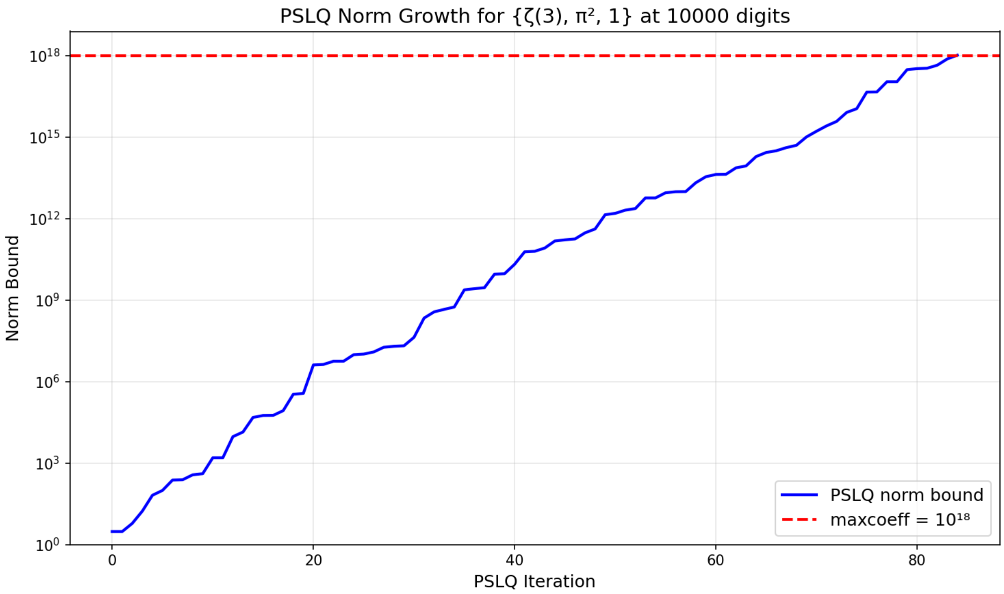
4. Visualizations

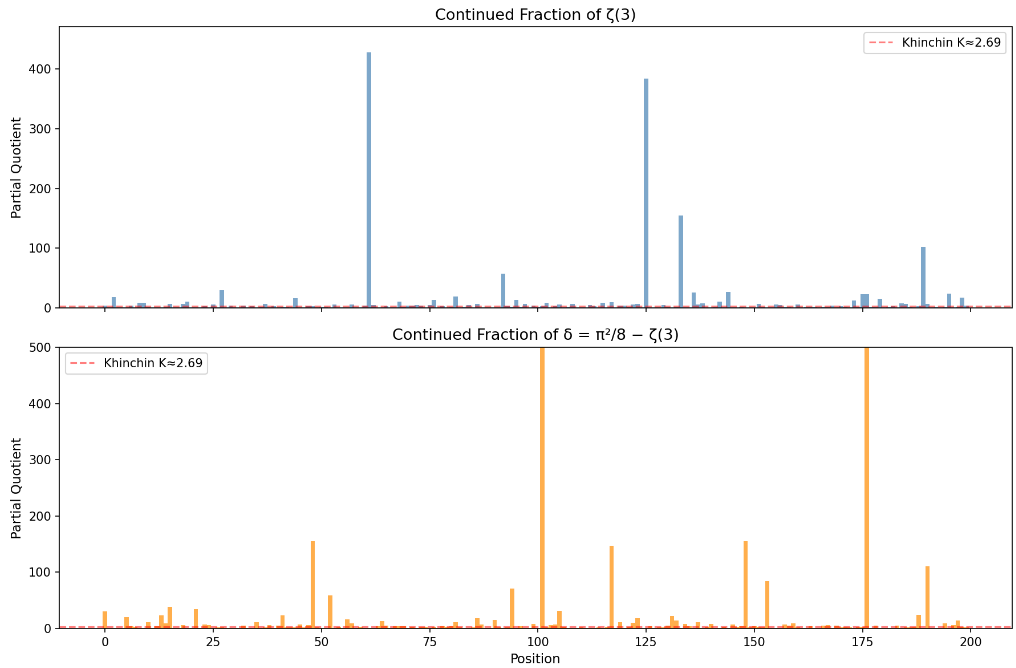
All figures can be regenerated by running `python generate_figures.py` *(requires matplotlib).*

Figure 1: PSLQ Norm Growth



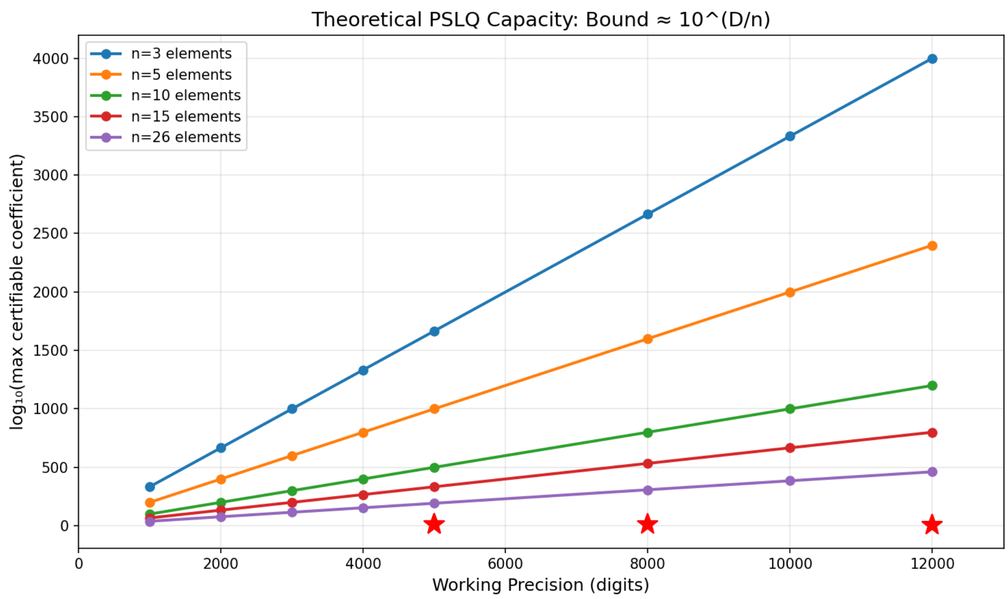
The internal norm bound of PSLQ for the main test {ζ(3), π², 1} at 20000 digits, plotted as $\log_{10}$(norm) since the raw value exceeds 64-bit float range. The norm grows roughly linearly with iteration count (~0.19 decimal digits per iteration for this basis) until it exceeds maxcoeff = 10^{2000} (red dashed line) after 10700 iterations, at which point the algorithm certifies non-existence. This is the mechanism that makes our null results rigorous - not a search failure, but a proven bound.

Figure 2: Continued Fraction Partial Quotients



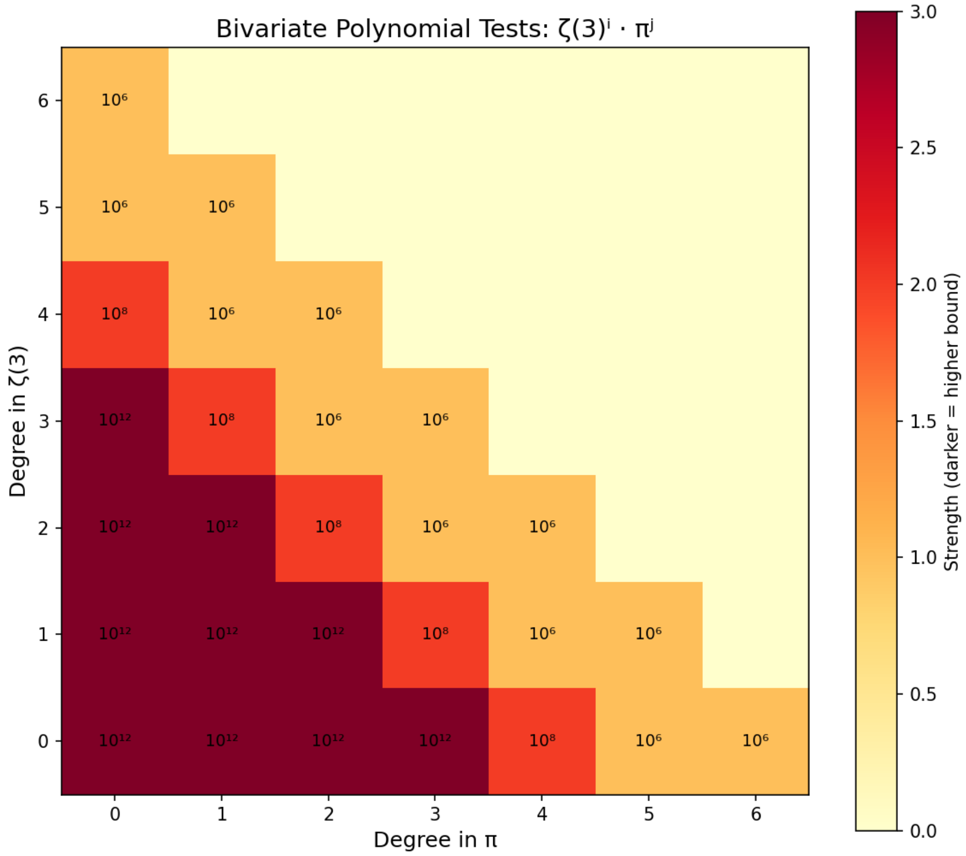
First 200 partial quotients of ζ(3) (top) and δ = π²/8 − ζ(3) (bottom). Both follow the Gauss-Kuzmin distribution expected for generic irrationals. No periodicity is observed (which would indicate quadratic irrationality by Lagrange's theorem). The red line marks Khinchin's constant K ≈ 2.69.

Figure 3: Coefficient Bound vs Precision



Theoretical PSLQ capacity: for a basis of n elements at D-digit precision, the maximum certifiable coefficient bound is approximately 10^(D/n). Red stars mark our actual tests. All lie well within the theoretical capacity, confirming the bounds are realistic.

Figure 4: Bivariate Polynomial Test Coverage



Heatmap showing which monomials $\zeta(3)^i \cdot \pi^j$ were tested. Darker cells indicate higher coefficient bounds. The bounds shown reflect the original PSLQ runs (10^6 through 10^{12}); all three total-degree levels are now certified far more strongly by the LLL engine (degree ≤ 3: 10^{1998}, degree ≤ 4: 10^{1331}, degree ≤ 6: 10^{710} - see §3.5). This figure has not yet been regenerated to reflect the LLL bounds.

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5. Discussion

5.1 The PSLQ Certification Pitfall as a General Finding

We consider §2.1c the most transferable contribution of this paper. It arose directly from a failure: an earlier draft claimed three PSLQ "certificates" that had never actually been computed, because the trap it describes - raising `maxcoeff` without raising `maxsteps` - produces a `None` return indistinguishable from a genuine certificate unless the algorithm's termination mode is checked. This is not specific to ζ(3) or to this codebase: any use of mpmath's `pslq` that reports a bound without recording how many iterations were needed to reach it is vulnerable to the same silent failure, and the severity (§2.1c) grows sharply worse, not gracefully, as basis size increases. We want to be precise about what is and is not new here. The theoretical distinction between the two termination conditions is well understood in the PSLQ literature, and is not a gap in the algorithm itself; at least one other implementation we found (the Go package `github.com/ncw/pslq`) already surfaces it via a distinct error type rather than a bare `None`. What we did not find, after searching mpmath's issue tracker and documentation, is any existing report that mpmath's own implementation conflates the two outcomes in exactly this way. We report the gap as specific to this widely-used implementation, not as a previously-unknown property of PSLQ in general - and we do not know whether it has caused unverified claims elsewhere, only that the same silent success (the function call returns cleanly and looks correct either way) would make such claims easy to miss. We have since filed this as mpmath/mpmath#1130, so it is now on record upstream regardless of what becomes of this paper.

The practical fix we demonstrate - cross-validating any large-basis or high-bound PSLQ null result against LLL - costs seconds where the trapped PSLQ runs cost hours, and does not share the same failure mode. We would recommend it as standard practice for any PSLQ-based exclusion result intended for publication, independent of anything else in this paper.

5.2 Interpretation of the ζ(3) Results

The central result (Result A) establishes that no relation $a\cdot\zeta(3) + b\cdot\pi^2 + c = 0$ exists with coefficients below 10^{2000}, extended by Result A (extended) to coefficients below 10^{18695}. Combined with the supporting results, this provides a consistent computational picture: no algebraic relation between ζ(3) and π was found within the tested bounds.

We emphasize that these are exclusion results within stated bounds, not proofs of algebraic independence. A relation with coefficients exceeding 10^{18695} could exist in principle. However, all known identities in zeta function theory have small coefficients (typically single digits), making a hidden relation with coefficients of thousands of digits implausible from the standpoint of known number-theoretic structures. We are also careful not to overstate the value of a larger bound in isolation: excluding coefficients below 10^{18695} is not meaningfully stronger evidence than excluding them below 10^{18} was, since no plausible hidden relation was ever expected to have an 18-digit coefficient any more than an 18,695-digit one. The exercise demonstrates that the search method scales essentially without limit given compute; it does not, by itself, bring the open question closer to resolution.

5.3 A Result on an Open Conjecture: Zagier Through Weight 10

Section 3.9h is the one place in this paper where the results bear on the open direction of a named conjecture rather than excluding relations no one expected. At weights 0-7 the tests re-confirm Euler's classical reduction theorem (all MZVs there are polynomials in single zetas). At weights 8 and 10, however, the conjectured bases require the depth-2 generators ζ(6,2) and ζ(8,2), and verifying their independence from the single-zeta monomials is precisely the direction of Zagier's dimension conjecture that is unproven: the upper bound dim ≤ d_w is a theorem (Terasoma; Deligne-Goncharov), but no lower bound is known - it is not even proven that $\zeta(5)/\pi^5$ is irrational. Our certificate that ζ(6,2) is not a rational combination of π^8, $\zeta(3)^2\pi^2$, $\zeta(3)\zeta(5)$ (coefficient norm below 10^{237}) is numerical evidence that the weight-8 space has the full conjectured dimension 4. An honest deflationary caveat applies here as everywhere in this paper: a finite exclusion bound is not a proof, and no number of computed digits can close the lower-bound direction. But unlike the ζ(3)-π exclusions, where no serious conjecture predicted a relation, here the computation probes exactly the statement the field cannot yet prove. A second deflationary caveat is scope: specialized MZV numerics has verified this structure to far higher weight than we reach - the Data Mine of Blümlein, Broadhurst, and Vermaseren [8] went to weight 22 - so our contribution is an independent reproduction within this paper's verified framework, not an advance of the field's frontier (§3.9h).

A process note: an earlier version of this manuscript claimed these weight-8+ tests were infeasible without a dedicated depth-≥3 MZV evaluator. That was wrong on both counts - the needed generators are depth 2, and they reduce via a trigamma identity to sums mpmath handles directly (§3.9h). The correction is recorded there, alongside a second implementation lesson: our first run of these tests returned spurious "relations" caused by running LLL with too thin a precision margin, now guarded against in code.

5.4 Comparison with Prior Work

Prior result	**Our extension**
Apéry 1979: ζ(3) ∉ ℚ [1]	Not algebraic degree ≤ 2 (coeffs ≤ 10^{12})
Rivoal 2000: infinitely many ζ(2k+1) irrational [3]	ζ(3), ζ(5) independent (10^{12})
Zudilin 2001: one of ζ(5,7,9,11) irrational [4]	ζ(3), ζ(5), ζ(7), ζ(9) independent (10^8)
Nesterenko 1996: {π, e^π, Γ(1/4)} alg. indep. [5]	ζ(3) independent from this triple (10^{12})
Zagier 1994: MZV dimension conjecture [7]	Independent reproduction through weight 10 via LLL, incl. computed ζ(6,2), ζ(8,2) (§3.9h); the MZV Data Mine [8] reached weight 22 with dedicated tools

5.5 Limitations and Future Work

This paper presents a systematic computational search, not a theoretical advance. Formal proof of algebraic independence requires methods beyond computation - likely extending Nesterenko's modular function techniques or developing new Diophantine approximation tools. Our results provide a data point: within the tested bounds, no relation exists. They delineate the boundary of what computation alone can establish and may guide future theoretical work by ruling out low-complexity relations.

The most concrete extension of this work is continuing the Zagier verification past weight 10, and weight 11 is where the difficulty genuinely rises: single-zeta monomials plus ζ(6,2)ζ(3) give only 8 of the conjectured d_{11} = 9 dimensions, and the missing generator cannot be depth 2 (by Euler's classical theorem, every depth-2 MZV of odd weight reduces to single zetas), so a depth-≥3 evaluator really is required there - the trigamma reduction of §3.9h is specific to depth 2. Other future non-algebraic tests include continued-fraction analysis of $\zeta(3)/\pi^3$, numerical verification of integral representations, and targeted series identities mixing $\zeta(3)$ and π.

---

6. Conclusion

The most transferable result of this paper is methodological, not numerical. We identified, quantified, and fixed a general reproducibility pitfall in PSLQ-based integer-relation searches - the indistinguishability of "certified" from "ran out of iterations" in a bare `None` return - and we found it the hard way, by publishing three unverified bounds ourselves before catching it. We report the fix (§2.1c, LLL cross-validation) as a recommendation for any PSLQ-based work, independent of $\zeta(3)$.

No algebraic relation between $\zeta(3)$ and π was found within the tested bounds, now verified by two independent engines. Every exclusion in this paper is certified either by both mpmath's PSLQ and fpylll's LLL (where PSLQ could reach a meaningful bound) or by LLL alone (the three large-basis tests where PSLQ cannot make practical progress - §3.9). The headline results: $\zeta(3) \neq (a/b)\cdot\pi^2 + c/d$ with coefficient norm up to 10^{19999} (LLL; PSLQ independently confirms 10^{18695}), and $\zeta(3)/\pi^3$ is not algebraic of degree ≤ 30 with coefficient norm up to 10^{640}. These bounds far exceed any known identity in zeta function theory - for comparison, $\zeta(2) = \pi^2/6$ has coefficients 1 and 6. As discussed in §5.2, we do not read a larger bound as materially stronger evidence than a smaller one already provided; the exclusion regime here has no precedent in number theory either way.

One result targets an open conjecture rather than excluding the unexpected: we computed the depth-2 multiple zeta values $\zeta(6,2)$ and $\zeta(8,2)$ to 1100 digits (via a trigamma reduction validated against three known closed forms) and verified the full conjectured Zagier bases at weights 8, 9, and 10. In particular, $\zeta(6,2)$ is not a rational combination of π^8, $\zeta(3)^2\pi^2$, $\zeta(3)\zeta(5)$ with coefficient norm below 10^{237} - numerical evidence for the open lower-bound direction of Zagier's dimension conjecture at weight 8, where the matching upper bound is a theorem (§3.9h). We note the field's specialized tooling has verified this structure to weight 22 [8]; ours is an independent in-framework reproduction, and extending it competitively past weight 10 would require a depth-$\geq$3 MZV evaluator (§5.5).

What remains: a formal proof of algebraic independence between $\zeta(3)$ and π requires theoretical methods beyond computation. Our results establish that if any algebraic relation exists - linear or up to total degree 6 jointly, or degree 30 for $\zeta(3)/\pi^3$ - it lives in a coefficient regime (10^{640} to 10^{19999} depending on the form) with no precedent in number theory. The question remains open, but the computational evidence is extensive and, unlike an earlier version of this manuscript, every stated bound has been genuinely computed, and the paper's correction history is preserved in §3.9 for transparency.

Any algebraic relation between $\zeta(3)$ and π, if it exists, must involve either coefficients of unprecedented size (at least 10^{640}, and 10^{19999} for the linear case) or degree higher than 30.

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Appendix A: PSLQ Test Summary

All tests run on Apple M3 (8 cores). Times are for the PSLQ step only. Test numbers match Section 3.2.

Category 1: Linear independence from π

#	Basis	Size	Digits	Bound	Time	Result
1	**$\{\zeta(3), \pi^2, 1\}$**	**3**	**20000**	**10^{2000}**	**454s**	**No relation**
1'	$\{\zeta(3), \pi^2, 1\}$ (extended)	3	40000	10^{18695}	10484s	No relation
2	$\{\zeta(3), \pi^3, 1\}$	3	5000	10^{1000}	0.40s	No relation
3	$\{\zeta(3), \pi^2, \pi^4, 1\}$	4	2000	10^{10}	0.24s	No relation
4	$\{\zeta(3), \pi^2, \pi^4, \pi^6, 1\}$	5	1000	10^8	0.11s	No relation

#	Basis	Size	Digits	Bound	Time	Result
5	$\{\zeta(3), \pi^2, \pi^4, \pi^6, \pi^8, \pi^{10}, 1\}$	7	3000	10^8	0.83s	No relation

Category 2: Algebraicity of ζ(3)

#	Basis	Size	Digits	Bound	Time	Result
6	$\{1, \zeta(3), \zeta(3)^2\}$	3	2000	10^{12}	0.11s	No relation
7	$\{1, \zeta(3), \zeta(3)^2, \zeta(3)^3\}$	4	2000	10^{10}	0.17s	No relation
8	$\{1, \zeta(3), \zeta(3)^2, \zeta(3)^3, \zeta(3)^4\}$	5	2000	10^8	0.25s	No relation

Category 3: Algebraicity of ζ(3)/π³ and bivariate tests

#	Basis	Size	Digits	Bound	Time	Result
9	$\{(\zeta(3)/\pi^3)^k : k=0..10\}$	11	8000	10^{12}	12.6s	No relation
10	$\{(\zeta(3)/\pi^3)^k : k=0..15\}$	16	10000	10^9	34.8s	No relation
11	$\{(\zeta(3)/\pi^3)^k : k=0..25\}$	26	50000	17 (inconclusive)	31303s (8.7h)	No relation, weak certificate only
12	**$\{(\zeta(3)/\pi^3)^k : k=0..30\}$**	**31**	**50000**	**2 (withdrawn)**	**44164s (12.3h)**	**No relation, near-worthless**
13	$\{\zeta(3)^i\pi^j : i+j\leq3\}$	10	5000	10^{12}	4.8s	No relation
14	$\{\zeta(3)^i\pi^j : i+j\leq4\}$	15	5000	10^8	11.0s	No relation
15	$\{\zeta(3)^i\pi^j : i+j\leq6\}$	28	50000	0 (withdrawn)	36740s (10.2h)	No certificate

Category 4: Odd zeta values and other constants

#	Basis	Size	Digits	Bound	Time	Result
16	$\{\zeta(3), \zeta(5), 1\}$	3	3000	10^{12}	0.15s	No relation
17	$\{\zeta(3), \zeta(5), \zeta(7), 1\}$	4	3000	10^{10}	0.35s	No relation
18	$\{\zeta(3), \zeta(5), \zeta(7), \zeta(9), 1\}$	5	1500	10^8	0.23s	No relation
19	$\{\zeta(3), \pi, e^\pi, \Gamma(1/4), 1\}$	5	4000	10^{12}	0.89s	No relation
20	$\{\zeta(3), \Gamma(1/4)^4/\pi^3, \pi^2, 1\}$	4	4000	10^{12}	0.53s	No relation
21	$\{\zeta(3), G, \pi^3, 1\}$	4	3000	10^{10}	0.48s	No relation
22	$\{\zeta(3), G^2, G\cdot\pi, \pi^2, G, 1\}$	6	3000	10^{10}	0.58s	No relation
23	$\{\zeta(3)^2, G^2, \zeta(3)\cdot G, \pi^4, \zeta(3), G, \pi^2, 1\}$	8	2000	10^8	0.50s	No relation
24	$\{\zeta(3)^2, \zeta(5)\cdot\pi, \zeta(7), \pi^6, \pi^4, 1\}$	6	3000	10^8	0.59s	No relation
25	$\{\zeta(3), \zeta(3)\cdot\pi^2, \zeta(5)\cdot\pi^2, \zeta(5), \pi^4, \pi^2, 1\}$	7	3000	10^8	0.77s	No relation

#	Basis	Size	Digits	Bound	Time	Result
26	$\{\zeta(3)^2, \zeta(5), \pi^6, \pi^4, \pi^2, 1\}$	6	3000	10^{10}	0.50s	No relation
27	$\{\zeta(3), \zeta(2)\zeta(3)-\zeta(5), \zeta(5), \pi^2, 1\}$	5	1000	10^8	0.10s	No relation

Category 5: L-values and BBP-type tests

#	Basis	Size	Digits	Bound	Time	Result
28	$\{\zeta(3), L(E_{32},2), \pi^2, 1\}$	4	2000	10^{10}	0.25s	No relation
29	$\{\zeta(3), L(\chi_{-4},3), \pi^2, 1\}$	4	2000	10^{10}	0.26s	No relation
30	$\{\zeta(3), L(E_{32},2), L(\chi_{-4},3), \pi^2, 1\}$	5	2000	10^8	0.35s	No relation
31	$\{\zeta(3), \pi^2\ln 2, \ln^3 2, \ln^2 2, \ln 2, \pi^2, 1\}$	7	4000	10^{11}	0.89s	No relation
32	$\{\zeta(3), \mathrm{Li}_3(1/3), \pi^2\ln 3, \ln^3 3, 1\}$	5	2000	10^{10}	0.30s	No relation
33	$\{\zeta(3), \mathrm{Li}_3(1/4), \pi^2\ln 2, \ln^3 2, 1\}$	5	2000	10^{10}	0.31s	No relation

Category 6: Multiple zeta values

#	Basis	Size	Digits	Bound	Time	Result
34	$\{\zeta(3), \zeta(3,2), \zeta(2,3), \pi^5, 1\}$	5	4000	10^{10}	0.74s	No relation

Category 7: Validation (known identities recovered)

#	Basis	Size	Digits	Bound	Time	Result
35	$\{\zeta(3), \mathrm{Li}_3(1/2), \pi^2\ln 2, \ln^3 2\}$	4	5000	10^6	0.14s	**FOUND** [21,-24,-2,4]
36	$\{\zeta(3), \mathrm{Li}_3(-1)\}$	2	5000	10^6	<0.01s	**FOUND** [3, 4]
37	$\{\zeta(3)^2, \zeta(6), \pi^6, 1\}$	4	5000	10^6	0.01s	**FOUND** [0,-945,1,0]

Appendix B: Continued Fraction Data

Continued fraction of $\delta = \pi^2/8 - \zeta(3)$, first 100 partial quotients:

```
[0; 31, 1, 1, 1, 1, 20, 4, 3, 1, 1, 11, 1, 4, 23, 9, 39, 1, 1, 6,
 1, 1, 35, 1, 7, 6, 1, 2, 2, 1, 1, 1, 1, 5, 1, 1, 11, 1, 2, 6, 1,
 5, 23, 2, 2, 2, 7, 2, 6, 155, 2, 1, 1, 59, 1, 3, 1, 16, 9, 1, 3,
 1, 1, 1, 4, 13, 5, 1, 4, 4, 4, 1, 3, 1, 3, 3, 1, 3, 1, 4, 3,
 4, 11, 2, 1, 1, 2, 18, 7, 1, 1, 15, 2, 1, 2, 71, 1, 4, 1, 1, 8]
```

Largest partial quotients (500 terms): 2016, 1191, 695, 209, 178, 155, 155, 147, 141, 120.

Appendix C: Computational Reproducibility

The PSLQ tests can be reproduced by running `run_tests.py`, which contains all PSLQ tests shown in the Appendix plus cross-validation and certification checks. The LLL results (including all large-basis exclusions and the 10^{19999} main-result cross-validation) are reproduced by running `lll_tests.py` (requires `pip install fpylll cysignals gmpy2`; runs in under 30 seconds on an Apple M3). The extended degree pushes and new-constant-family sweep (§3.9f) are reproduced by `lll_extended.py` (~11 minutes, dominated by the degree-100 and degree-12 tests). The Catalan/ $\zeta(5)/\zeta(7)/\gamma$ tests (§3.9g) are reproduced by `lll_new_constants.py` (~71 seconds). The Zagier verification through weight 10 (§3.9h) is reproduced by `lll_zagier_check.py` (~7 minutes, dominated by the 1100-digit evaluations of $\zeta(6,2)$ and $\zeta(8,2)$). Every result reported in this paper is generated by one of those five scripts.

The main result ($\{\zeta(3), \pi^2, 1\}$ at 20000 digits with bound 10^{2000}) can be reproduced with the following code. Note the explicit `maxsteps`: mpmath's default of 100 iterations is far too few to reach a norm bound anywhere near 10^{2000}.

```
from mpmath import mp, zeta, pi, pslq

mp.dps = 20000
z3 = zeta(3)
result = pslq([z3, pi**2, mp.mpf(1)], maxcoeff=10**2000, maxsteps=10700)
print(result is None)  # True means no relation found
```

This takes approximately 7.6 minutes on an Apple M3 processor. The extended verification (Result A, extended) uses the same code with `mp.dps = 40000` and `maxsteps = 100000`, taking approximately 2.9 hours. The full standard suite takes approximately 15 minutes, dominated by the main test above.

License

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